Do Vision-Language Models Track Visual State? A Controlled Study of Trajectory Supervision in Qwen2.5-VL

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Abstract

Does training a vision-language model to write out intermediate states teach it to track an object through a video? We test this on a card-relocation task (50 videos, four positions, six rounds) with Qwen2.5-VL-3B-Instruct and QLoRA. We compare the zero-shot Base model, trajectory supervision (B), and a format-matched negative control (B′v3) whose training traces are borrowed from other videos but match B in structure, move counts and letter frequencies. Under 5-fold cross-validation over all 50 videos, B beats Base on per-round accuracy (0.350 vs. 0.200), but B′v3 recovers most of that gain (0.323). The difference between B and B′v3 is small and not significant ($+0.027$, 95% CI $[0.000, +0.067]$, $p = 0.25$). A diagnostic explains why: B's per-round accuracy equals a policy that always repeats the starting position (0.35), and B writes one constant letter for all six rounds in 42 of 50 videos. On rounds where the target has moved, all three models are near zero (B 3%, B′v3 7%, Base 7%), final-position accuracy is at chance, and no model produces a fully correct trace (0/50). Aggregate per-round accuracy can reward a model that tracks nothing, so we recommend reporting a stay-put baseline and moved-away accuracy next to it.

1. Introduction

Chain-of-thought prompting made language models better at multi-step problems by having them write out intermediate steps [1]. A natural idea for video is to do the same with state: train a vision-language model (VLM) to write down where an object is after every event, and hope the written trace forces it to track the object. There is a competing explanation. The model may only learn the output format, and a trace that looks right says nothing about whether the video was used.

We test this on a deliberately narrow task. Four cards sit at four fixed positions (A to D), one card is the target, and six rounds of swaps move it around. The ground truth after each round is a single letter, so scoring is unambiguous. We fine-tune Qwen2.5-VL-3B-Instruct [4] with QLoRA [6] and compare three conditions: the untouched Base model, a model trained on the true six-round trajectory (B), and a negative control (B′v3) trained on traces with the same format and statistics but with the middle rounds borrowed from other videos, so the labels no longer match the video.

An earlier version of this study used a ten-video test set, which was too small to support any conclusion. Here we run 5-fold cross-validation over all 50 videos, so every video is tested once by a model that never trained on it, and we compare conditions with paired video-level tests. The main results are:

- B improves per-round accuracy over Base (0.350 vs. 0.200), but the format-matched control recovers most of the gain (0.323). B minus B′v3 is $+0.027$ (95% CI $[0.000, +0.067]$, $p = 0.25$).
- The gain is what a no-tracking policy would score. Always repeating the starting position gets 0.35. B writes one constant letter for all six rounds in 42 of 50 videos.
- Where the target has moved away from its start, every condition is near zero (B 3%, B′v3 7%, Base 7%). Final-position accuracy is at chance, and no condition produces a fully correct trace (0/50).

Contributions.

1. A format-matched, corrupted-trajectory control for trajectory supervision, built by move-count-matched cross-example assignment and rebuilt for every fold.
2. A 5-fold, 50-video evaluation with paired video-level statistics, which gives a null result for B vs. B′v3 with a bound on how large a missed effect could be.
3. A stay-put baseline and a moved-away accuracy metric, which show that the apparent gain over zero-shot is not tracking.

2. Related Work

VLMs for video. Qwen2-VL [3] and Qwen2.5-VL [4] handle images and video through a shared pathway. Broad

benchmarks such as Video-MME [5] cover many abilities at once. We use one narrow task so that a single ability, updating a state across rounds, can be isolated.

Intermediate supervision. Chain-of-thought prompting [1] elicits intermediate steps from language models. We ask whether training on intermediate *visual* states does more than teach a format. The corrupted-label control is our way of separating the two.

Shortcut learning. Models often exploit regularities in the data instead of the intended solution [2]. Two shortcuts matter here: predicting that the target never moves, and copying the starting position from the prompt (Section 3.1).

3. Task, Data and Setup

3.1. Task and data

We use the card-relocation subset of the *video-mme-logical* dataset (Hugging Face: `marcuskwan/video-mme-logical`), 50 videos. Four cards occupy four fixed positions A to D, one card is the target, and the video shows six rounds of swaps. The ground truth is the target's position after each round. Each video is read as 24 frames at 224×224 through the native Qwen video path.

In all 50 videos the target card's label equals its starting position, so the prompt reveals the initial position. Accuracy on the initial position (about 76% in our early diagnostic) therefore says nothing about visual grounding. We score only rounds 1 to 6 and the final position. The target is at its start in 106 of the 300 (video, round) pairs and away from it in the other 194. A policy that always outputs the starting position scores 0.35 per round and 0.28 on final position.

3.2. Cross-validation

We use five folds of ten videos. Fold 0 is the original fixed held-out set (seed 20260917). The remaining 40 videos are dealt into folds 1 to 4, stratified by final position (seed 20261001). For each fold, models train on the other 40 videos and are tested on the ten held-out ones, so each video is tested exactly once by a model that did not see it. The Base model is evaluated once on all 50 videos. Fold 0 reuses the adapters from the original fixed-split run, and folds 1 to 4 use the same recipe. The B′v3 data are rebuilt from each fold's own 40 training videos.

3.3. Model and training

We fine-tune Qwen2.5-VL-3B-Instruct [4] with QLoRA [6]: 4-bit NF4 with double quantization, rank 16, alpha 32, dropout 0.05, adapters on the language model's attention and MLP layers. The vision tower and the visual-text merger are frozen. Batch size is 1 with 8 gradient-accumulation steps, learning rate 10^{-4} with a cosine schedule, weight decay 0.01, 8-bit AdamW, 3 epochs (15 optimizer steps), maximum sequence length 4096, fp16 on a single T4. Decoding is greedy, with at most 16 new tokens for the final-answer prompt and 256 for the trajectory prompt. We evaluate one training configuration.

3.4. Conditions

Base. The pretrained model with no fine-tuning, in the same 4-bit pipeline.

B (trajectory). The prompt names the target card and asks for the position after every round in a fixed format (`Initial`, `Round 1` to `Round 6`, `Final`). The training target is the true trace.

B′v3 (corrupted control). Same prompt, same output format. The `Initial` and `Final` labels (and round 6) stay correct. Rounds 1 to 5 are replaced by the rounds of a different training video. The source is chosen by the Hungarian algorithm [7]: for video v_i the cost of borrowing from v_j ($j \neq i$) is $|\text{moves}(v_i) - \text{moves}(v_i \text{ with } v_j\text{'s rounds})|$, so the move-count distribution is preserved. In the original split this gave a letter-frequency L1 distance of 0, mean move count 2.975 (true) vs. 3.000 (control), a move-histogram L1 distance of 2, no identical traces, and a majority of intermediate rounds changed in 31 of 40 videos. Per-fold audits are saved with the code.

Two other conditions (final-answer-only and transition supervision) were trained in the earlier ten-video stage. They are not part of the cross-validation and are not reported.

3.5. Evaluation and statistics

Each video gets three scores: per-round accuracy (fraction of six rounds correct), final-position accuracy (from the `Final` line for the trajectory prompt, the answer letter for the final-answer prompt), and exact match (all six rounds correct). Unparseable outputs count as wrong; there were none. The unit of analysis is the video ($n = 50$).

Confidence intervals are Clopper-Pearson [9] for binary scores and a bootstrap over videos (10,000 resamples) for per-round accuracy. Paired differences get a bootstrap interval and a sign-flip permutation test (100,000 flips) on per-video differences. Binary scores also get an exact McNemar test [8]. We designated per-round accuracy on the trajectory prompt, B vs. B′v3, as the primary comparison before running the folds. Moved-away accuracy was added after the stay-put diagnostic and is post hoc. We apply no multiplicity correction.

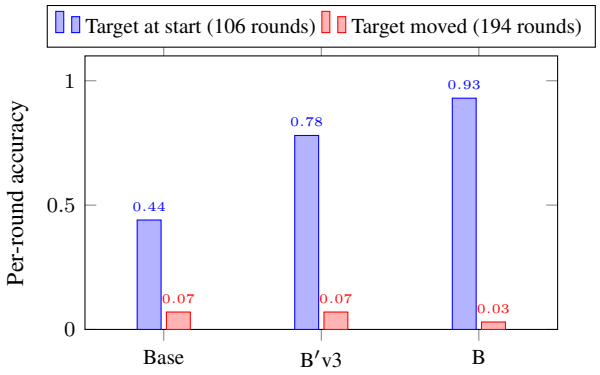


Figure 1. Per-round accuracy split by whether the target is at its starting position. Training raises accuracy only where the target has not moved.

4. Results

4.1. Aggregate accuracy

Table 1 shows that both trained models beat Base on per-round accuracy. Final-position accuracy is 0.24 to 0.26 for every condition, against 0.25 for random guessing and 0.30 for the majority letter, and no condition produces a fully correct trace. An earlier fp16 run of the Base model on the same 50 videos (native video input, 24 frames) gave the same picture: 0/50 exact matches and 12/50 correct final positions. Base given only the final frame also scored 12/50.

4.2. Paired comparisons

Table 2 gives the paired results. For the primary comparison, B minus B′v3 is $+0.027$ per round with an interval that touches zero ($p = 0.25$). The upper end of that interval is $+0.067$, so we can exclude an advantage of B over the control larger than about seven points of per-round accuracy. For final-position accuracy the intervals are about ± 0.13 wide, so only large differences are excluded. On the trajectory prompt the discordant pairs are 3 vs. 4. Both trained models beat Base per round ($p < 0.001$), and neither differs from Base on final position.

4.3. What the gain is

Table 3 and Figure 1 decompose the per-round numbers. B predicts the starting position in 93% of rounds, and in 42 of 50 videos it writes the same letter six times. Its per-round accuracy of 0.350 is the stay-put baseline. Essentially all of its gain over Base comes from rounds where the target is at its start (0.93 vs. 0.44). On the 194 rounds where the target has moved, B is correct on 6, B′v3 on 14 and Base on 13. A uniform guess over the four positions would score 0.25 on these rounds, so all three conditions are below chance.

B′v3 recovers roughly four fifths of B's gain over Base (0.123 of 0.150) while being trained on traces that do not match the video. It collapses less (78% stay-put), plausibly because its borrowed rounds contain moves. The post hoc moved-away difference between B and B′v3 (-0.045, $p = 0.031$) is the only uncorrected $p < 0.05$ in the study. It is the same effect seen from the other side: B answers "never moved" more often, so it has fewer chances to be right once the target moves. We do not claim that trajectory supervision harms tracking.

4.4. Consistency across folds

Table 4 shows that the sign of B minus B′v3 is not stable: positive in three folds, zero in one, negative in one, and no fold exceeds $+0.067$. This is what noise around a small true difference looks like.

5. Discussion

5.1. What the evidence supports

1. Trajectory supervision raises per-round accuracy over zero-shot (B $+0.150$, B′v3 $+0.123$).
2. That gain matches the stay-put baseline, and a corrupted-label control recovers most of it.
3. On rounds where the target has moved, accuracy is near zero in every condition, and training does not improve it.
4. Final-position accuracy is at chance for all conditions, and no model produces a fully correct trace (0/50).
5. Correct labels gave no detectable advantage over corrupted ones. B's stronger collapse toward stay-put is an exploratory observation, not a confirmed harm.

We do not claim that format learning "dominates" or that supervision on intermediate states cannot work. We claim that, in this setup, we could not detect tracking in any condition.

5.2. A possible mechanism

Because the target label equals its starting position, the first line of every trace (`Initial`) is determined by the prompt. Both B and B′v3 keep that line correct in training, and B's outputs mostly repeat it. This is consistent with anchor copying: the model learns to carry the prompt's letter through the trace. We have not tested this. Possible tests are removing the anchor from the prompt, evaluating on videos where the target does not start at its label, or probing the model's internal state.

5.3. Relation to the earlier ten-video study

The first version of this study reported differences of one to three videos on a ten-video test set, and interpreted some of them (for example B scoring below Base on the final-answer prompt). With 50 videos these differences are gone: final-position accuracy is 0.24 to 0.26 everywhere. We report none of the earlier ten-video comparisons, because they were within noise and we now treat them as such.

	Per-round	Final (traj. prompt)	Final (final prompt)	Exact 6/6
B	0.350 [0.287, 0.413]	0.240 [0.131, 0.382]	0.260 [0.146, 0.403]	0/50
B′v3	0.323 [0.260, 0.387]	0.260 [0.146, 0.403]	0.240 [0.131, 0.382]	0/50
Base	0.200 [0.153, 0.247]	0.240 [0.131, 0.382]	0.260 [0.146, 0.403]	0/50
Random	0.250	0.250	0.250	–
Majority final	–	0.300	0.300	–
Stay-put	0.350	0.280	0.280	–

Table 1. Results over all 50 videos (5-fold CV for B and B′v3). Means over videos with 95% intervals: bootstrap over videos for per-round accuracy, Clopper-Pearson for the binary scores. Upper 95% bound for 0/50 is 0.071. Stay-put always predicts the starting position.

Metric	B − B′v3	B − Base	B′v3 − Base
Per-round (primary)	$+0.027$ [0.000, $+0.067$], $p = 0.25$	$+0.150$ [$+0.103$, $+0.197$], $p < 0.001$	$+0.123$ [$+0.087$, $+0.163$], $p < 0.001$
Final, trajectory prompt	-0.020 [-0.120, $+0.080$], $p = 1.00$	0.000 [-0.060, $+0.060$], $p = 1.00$	$+0.020$ [-0.100, $+0.140$], $p = 1.00$
Final, final-answer prompt	$+0.020$ [0.000, $+0.060$], $p = 1.00$	0.000 [-0.160, $+0.160$], $p = 1.00$	-0.020 [-0.180, $+0.120$], $p = 1.00$
Moved-away rounds (post hoc)	-0.045 [-0.089, -0.011], $p = 0.031$	-0.021 [-0.074, $+0.035$], $p = 0.52$	$+0.023$ [-0.036, $+0.087$], $p = 0.49$

Table 2. Paired video-level differences ($n = 50$) with 95% bootstrap intervals. p-values: sign-flip permutation test for per-round and moved-away accuracy, exact McNemar test for the final-position scores. No multiplicity correction.

5.4. Limitations

One run per fold. We trained one model per condition per fold, with no repeated seeds. The intervals reflect variation across 50 videos and five training sets, not seed-to-seed variance.

Small training set and one configuration. Each model sees 40 videos for 3 epochs (15 optimizer steps). A null result may reflect undertraining, and we evaluate a single configuration.

One small model, one synthetic task. Only Qwen2.5-VL-3B-Instruct with a frozen vision tower and language-side adapters, and only card relocation. Larger models, an unfrozen vision side or natural videos may behave differently.

Post hoc and uncorrected analysis. Moved-away accuracy was chosen after seeing the stay-put diagnostic, and no test is corrected for multiple comparisons.

No temporal-order control. An earlier frame-order shuffle used separate-image input and was invalid, so we make no claim about whether the models use temporal order.

Dataset property. The target label equals its starting position in all 50 videos, which makes the anchor-copying shortcut available. We have not checked how this subset was built.

Fold 0. Fold 0 reuses adapters from the earlier fixed-split run, trained with the same recipe.

6. Conclusion

On a controlled card-relocation task, trajectory supervision of Qwen2.5-VL-3B does not teach state tracking that we can detect. Per-round accuracy rises from 0.200 to 0.350, but that equals a policy that always repeats the starting position, a control trained on mismatched traces recovers most of it, and accuracy on rounds where the target has moved stays near zero. Final-position accuracy is at chance and no trace is fully correct. B vs. B′v3 is a null with an upper bound of about seven points of per-round accuracy.

The practical lesson is about reporting. A stay-put baseline and accuracy on rounds where the state has changed should sit next to any aggregate per-round number, because the aggregate can reward a model that tracks nothing. Replication with several seeds, larger models, longer training, a prompt without the starting-position anchor, and mechanistic probes would test whether the null holds. Fold assignments, raw model outputs, per-video scores and per-fold control audits are stored with the analysis code.

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	All-stay-put videos	Constant sequence	Pred. = initial (of 300 rounds)	Acc. target at start ($n = 106$ rounds)	Acc. target moved ($n = 194$ rounds)
B	42/50	42/50	0.93	0.93	0.031 [0.011, 0.066]
B′v3	32/50	33/50	0.78	0.78	0.072 [0.040, 0.118]
Base	0/50	0/50	0.28	0.44	0.067 [0.036, 0.112]

Table 3. Decomposition of per-round accuracy on the trajectory prompt. "All-stay-put" means every round equals the starting position. Moved-away intervals are Clopper-Pearson over pooled rounds, which treats rounds as independent and is optimistic; the video-level paired tests in Table 2 are the ones to cite.

Fold	B	B′v3	Diff	B fin.	B′v3 fin.
0	0.250	0.183	+0.067	0.20	0.20
1	0.267	0.233	+0.033	0.20	0.30
2	0.400	0.350	+0.050	0.40	0.40
3	0.383	0.400	−0.017	0.20	0.20
4	0.450	0.450	0.000	0.20	0.20

Table 4. Per fold ($n = 10$ videos each), trajectory prompt: per-round accuracy for B and B′v3, their difference, and final-position accuracy.

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